INCH-POUND
MIL-STD-419F
15 September 2021
SUPERSEDING
MIL-STD-419E
W/CHANGE 1
31 August 2017

DEPARTMENT OF DEFENSE STANDARD PRACTICE

CLEANING, PROTECTING, AND TESTING PIPING, TUBING, AND FITTINGS FOR HYDRAULIC POWER TRANSMISSION EQUIPMENT

AMSC N/A FSC 4730

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FOREWORD

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1. SCOPE

1.1 Scope. This standard covers the requirements governing the basic methods of cleaning and protecting nonferrous and ferrous metal and alloyed pipe, tubing, and fittings prior to installation and after hydraulic tests in hydraulic power transmission applications. This standard is applicable to hydraulic components used in systems filled with either petroleum oil, synthetic poly-alpha-olephin (PAO) fluid, phosphate ester type fluid, or water-based fluid.

1.2 Applicability. The procedures described in this standard are intended for the use of shipyard or contractor personnel only in accordance with the safety precautions of section 5.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERICAL ITEM DESCRIPTIONS

A-A-59441 - Vapor Corrosion Inhibitors

A-A-59563 - Sodium Carbonate, Anhydrous, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680 - Degreasing Solvent

MIL-PRF-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Operation

MIL-PRF-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application

MIL-DTL-16232 - Phosphate Coating, Heavy, Manganese or Zinc Base

MIL-D-16791 - Detergents, General Purpose (Liquid, Nonionic)

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

MIL-PRF-17672 - Hydraulic Fluid, Petroleum, Inhibited

MIL-H-19457 - Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic

MIL-H-22072 - Hydraulic Fluid, Catapult, NATO Code Number H-579

MIL-DTL-32353 - Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base

MIL-DTL-32594 - Sodium Phosphate, Tribasic; Anhydrous, Dodecahydrate, and Monohydrate, Technical

MIL-PRF-83282 - Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2073-1 - Standard Practice for Military Packaging

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-407 - Contamination Control Technology Precision Cleaning Methods and Procedures

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D6304 - Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration

(Copies of this document are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4406 - Hydraulic Fluid Power – Fluids – Method for Coding the Level of Contamination by Solid Particles

ISO 11171 - Hydraulic Fluid Power – Calibration of Automatic Particle Counters for Liquids

ISO 23309 - Hydraulic Fluid Power Systems – Assembled Systems – Methods of Cleaning Lines by Flushing

(Copies of these documents are available online at www.iso.org.)

NATIONAL AEROSPACE STANDARD (NAS)

NAS 847 - Caps and Plugs, Protective, Dust and Moisture Seal

(Copies of this document are available online at www.aia-aerospace.org.)

SAE INTERNATIONAL

SAE AS4059 - Aerospace Fluid Power – Cleanliness Classification for Hydraulic Fluids

SAE AS598 - Aerospace Microscopic Sizing and Counting of Particulate Contamination for Fluid Power Systems

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Beta ratio. The beta ratio is a measure of filtration performance equal to the average number of particles upstream of a filter divided by the average number of particles downstream of a filter for a specified particle size.

3.2 Descaling. Descaling is the treatment of metallic surfaces in order to remove impurities, stains, rust, or scale. It is performed prior to subsequent processing (such as extrusion, rolling, painting, galvanizing, or plating) or when necessary to meet cleanliness requirements prior to assembly or installation. Descaling can be either chemical, mechanical, or a combination of both. Chemical descaling, also commonly referred to as pickling, utilizes a solution called pickle liquor (typically containing strong mineral acids) to remove impurities. Mechanical descaling utilizes abrasive blast cleaning to remove impurities.

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3.3 Flushing. Flushing is the process of cleaning a hydraulic piping system that involves the circulation of turbulent flushing fluid within piping system loops to remove, transport, and filter out particles and other contaminants that may have been introduced into the system during manufacture, construction, or maintenance.

3.4 Outflushing. Outflushing is the unrestricted discharge of sufficient fluid volume to an open container or bucket to remove contamination from a dead end in the piping.

3.5 Passivation. Passivation is the final treatment/cleaning process of metallic surfaces to remove unwanted impurities (e.g., anodic contaminants) and to provide a uniformly noble-cathodic corrosion-resistant surface.

3.6 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

4. GENERAL REQUIREMENTS

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4.2 Cleanliness. Pipe tubing and fittings related to the hydraulic power transmission equipment shall be thoroughly cleaned and descaled as necessary prior to installation (see 5.2 and 5.3). They shall be free from scale and foreign matter which could be detrimental to operation of hydraulic equipment such as pumps, motors, valves, rams, and accumulators.

4.3 Pipe construction and cleaning. Pipe required to complete a hydraulic installation shall be constructed in a shop where facilities exist. However, shipboard or vehicle construction is allowed where provisions exist to allow cleanliness controls and preclude damage to equipment. Subassemblies shall be used to the maximum extent practicable. After construction and bending operations are completed, the constructed and bent pipe, tubing, and fittings shall be cleaned and descaled in accordance with the procedures specified herein (see 5.2 and 5.3). Hydraulic components (which may be constructed of materials not compatible with the chemical descaling solution) shall not be immersed in the cleaning or chemical descaling baths. Complex components, such as pumps, valves, and rams (which may be compatible with the chemical descaling solution but will require disassembly to remove residual fluids) shall not be immersed in the cleaning or chemical descaling baths. No hydraulic component that is mechanically connected (union, flange) to rigid piping shall be subject to the chemical descaling solution.

4.4 Abrasive blasting. Sand, shot, or other abrasive blasting shall not be permitted on any part of the hydraulic system. Unless a separate isolated room is provided in the shop for construction of hydraulic piping, abrasive blasting shall be prohibited in the same building in which an open hydraulic component is housed. Abrasive blasting, chipping, or grinding shall not be permitted in the vicinity of an installation site for an open hydraulic system.

4.5 Sand packing. Hydraulic pipe or tubing shall not be packed with sand during the bending process.

4.6 Wire brushing. The pipe and tubing shall not be wire brushed after descaling.

4.7 Cloth and paper. Cloth or paper material shall not be used in drying operations, nor for capping open ends (see 5.7).

4.8 Corrosion protection. The interior and exterior of the ferrous pipe, tubing, and fittings (except corrosion-resistant steel) shall be treated as specified in 5.5 to prevent corrosion.

4.9 Capping. Openings shall be capped and sealed airtight immediately after the cleaning process with a cap or plug in accordance with NAS 847 (see 5.7). The openings shall remain capped until immediately before connecting to the equipment for which the pipe is intended.

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4.10 Heat treated piping and tubing. Pipe or tubing that has had heat applied at the site of installation to assist in forming shall be returned to the shop or other areas where facilities exist for recleaning, additional descaling (if necessary), and retesting.

5. DETAILED REQUIREMENTS

5.1 Safety precautions. The contractor or shipyard representative shall ensure that this cleaning is coordinated through the safety superintendent to provide personnel protection from acid and chemical hazards. In general, all chemicals involved are harmful to the eyes and skin. Personal contact with chemicals should be avoided. Obtain medical attention immediately if acid or caustic solution gets in the eyes. Face shields or goggles shall be worn by personnel involved in chemical cleaning. Breathing mists or vapors over mixing tanks shall be avoided. Protective equipment, including personal protective equipment for eyes, face, head, and extremities, protective clothing, respiratory devices, and protective shields and barriers, shall be required for personnel who are mixing caustic or acidic cleaning solutions.

5.2 Cleaning. Cleaning and treatment compounds can be environmental pollutants if discharged to the water and may be toxic to freshwater and marine organisms. Disposal procedures specified in 5.9 shall be followed. Chemical descaling is not required to clean existing installations during repair or modification unless the pipe or tube has internal oxide scale that cannot be removed by wire brush, water jet, or flushing. If new piping sections have been installed, only the new pipe or tube requires descaling and passivation (see 5.3) prior to installation to remove rust, scale, and foreign matter. New piping sections do not require descaling and passivation if already performed prior to delivery and a clean condition maintained during construction. Heat discoloration of piping interior due to welding is not cause for chemical descaling. In preparation for descaling, surfaces coated with oil, preservatives, grease, or other visible contaminants shall be removed using an appropriate degreasing solution. An example of one such degreasing mixture is trisodium phosphate monohydrate in accordance with MIL-DTL-32594 (7 to 10 ounces [198 to 284 grams]), nonionic detergent polyethylene glycol monoalkylaryl ether in accordance with MIL-D-16791, type I (1 fluid ounce [29.6 milliliters]), and water (1 gallon [3.8 liters]), wire-brushed, and rinsed in accordance with a written procedure that has proven to be effective. If cleaned pipe, tube, and fittings are not descaled (see 5.3) immediately, dry the item and then protect it from dust, dirt, oil, and moisture.

5.2.1 Compressed air cleaning plugs. The use of commercially-available pipe cleaning plugs or “pigs” that are close fitting to the inside diameter of the pipe and driven by compressed air for mechanical removal of contaminants in corrosion-resistant pipe assembly is acceptable, provided that the pipe assembly contains no dead legs, branches, or reductions in area. Accountability shall be provided for any plugs that were used to ensure that they are not left behind in the pipe assembly.

5.3 Descaling and passivation. Chemical descaling, mechanical descaling, or any combination thereof necessary to accomplish removal of rust and scale from piping shall be performed in accordance with a written procedure that has proven to be effective. Excessive time in the chemical descaling solutions shall be avoided, as it could result in the degradation of the mechanical properties of the pipe, tubing, or fittings. The use of passivation treatments (such as an alkaline bath containing sodium carbonate in accordance with A-A-59563) following chemical or mechanical descaling or any combination thereof shall produce passivated surfaces for the purposes of maximizing corrosion resistance. The use of a descaling and passivation procedure shall result in piping provided in a descaled, passivated, and cleaned condition. Nonferrous and ferrous metal and alloyed pipe, tubing, and fittings shall be subject to descaling and passivation prior to hydrostatic testing (see 5.3.1).

5.3.1 Hydrostatic testing prior to installation. Piping assemblies shall be subjected to a hydrostatic test prior to installation at a test pressure, which is typically specified in system diagrams, drawings, or technical documentation. Hydrostatic testing shall be performed with tap water or hydraulic fluid equal to, or compatible with, that normally used in the system. If leaks occur, they shall be repaired and the assembly retested. Piping assemblies that have been subject to hydrostatic testing with hydraulic fluid shall be degreased just prior to additional brazing or welding. Following a satisfactory test, the solution shall be drained and all openings capped. If water is used, the assembly shall be blown dry by clean, dry air until no moisture is visible using any method that does not result in loss of cleanliness before openings are capped. The piping assembly shall then be installed up to the components (see 4.2).

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5.4 Flush of installed system for removal of brazing fluxes. If piping section(s) with brazing have been installed and the system has maintained cleanliness during repairs, only the sections of pipe or tube installed requires flushing in accordance with this section. After installation of the entire piping system, (up to but not including) the components shall be flushed to remove brazing fluxes. Flushing blocks and other jumpers shall be used to connect piping around components not being flushed. Before filling the system with water, a low pressure air test (up to 100 lb/in² [689.5 kPa]) shall be applied to the piping to check for leaks. Any one of the three following procedures shall be conducted. The hot flush and hot circulation methods are preferred.

a. Hot flush soak with non-recirculating fresh water for 1 hour while ensuring that the temperature at any part of the system does not go below 110 °F (43 °C).

b. As an alternative to the hot flush procedure, a hot recirculating procedure with fresh water may be conducted for a period of 1 hour for systems where such an arrangement is feasible. The system temperature shall be monitored so that no part of the system falls below 110 °F (43 °C). Following the recirculating procedure, the system shall be flushed with fresh water for 15 minutes at a minimum temperature of 60 °F (16 °C).

c. Cold soak the system for 12 hours using fresh water at a minimum temperature of 60 °F (16 °C). At the completion of the 12-hour soak, systems shall be flushed with nonrecirculating fresh water at a minimum temperature of 60 °F (16 °C) for 4 hours.

Under all of the above flux removal procedures, the system shall be full of water, so that joints are completely submerged at all times. The minimum flow rate in gallons per minute required for removal of residual brazing flux in piping systems shall be 1.5 times the pipe internal diameter in inches (the equivalent minimum flow rate in liters per minute shall be 2.2 times the pipe internal diameter in centimeters). Upon completion of the flux removal procedure, the system shall be completely drained and dried with clean air until no moisture is visible using any method that does not result in loss of cleanliness.

5.4.1 Requirements for final fit-up joints using minimum flux brazing procedure. Final fit-up joints that are used to install piping assemblies that were brazed and flushed prior to installation do not require a system flush following installation, provided that a qualified minimum flux brazing procedure is used for joint fit-up.

5.5 Hydrostatic test after installation. The entire piping system, after installation up to the components, shall be subjected to the specified test pressure with a hydraulic fluid equal to, or compatible with, the fluid normally used in the system. If a water-based hydraulic fluid has been used, the system shall be drained completely after the test and blown dry with clean air. Components and piping assemblies which passed a hydrostatic test prior to installation need not be retested.

5.5.1 Hydraulic system/component preservation. Hydraulic systems and components which are subject to corrosion shall be preserved in accordance with MIL-STD-2073-1. Prior to all assembly, all oil films and preservatives shall be removed from those sections which shall be subjected to welding or brazing operations. Assembled systems which have been protected by a preservative fluid (see 5.5.1.1) shall be thoroughly drained and blown down before adding the system fluid. See 5.9 for disposal instructions for these fluids. If the assembled system has not been thoroughly cleaned prior to adding protective fluids, a complete flush (see 5.6) shall be required before putting the system into operation.

5.5.1.1 Assembled systems. Hydraulic systems which are to be left inactive for 6 months or less shall be protected by filling with the system fluid. Those systems, fabricated from non-corrosion resistant piping, which are to be left inactive for an extended period (6 months or more) shall be filled with a preservative fluid. For systems containing petroleum base or PAO base fluids, the preservative fluid shall be in accordance with MIL-PRF-6083. For systems containing phosphate ester base hydraulic fluids (such as fluids in accordance with MIL-H-19457), the preservative fluid shall be in accordance with MIL-H-19457, with addition of a Vapor Corrosion Inhibitor (VCI) in accordance with A-A-59441, per the VCI manufacturer's instructions. For systems containing water-based hydraulic fluids, the preservative fluids shall be the system fluid in accordance with MIL-H-22072. Vents and tanks shall be provided to compensate for the expanding and contracting fluid and minor system leaks. When the system is preserved with other than the normal system fluid, tags with the following information shall be conspicuously attached: "This system has been filled with preservative fluid to (specification) on (date). This fluid shall be completely drained and blown down before adding system fluid."

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5.5.1.2 Components. If system components are not to be installed immediately, preservation shall be in accordance with the requirements specified in 5.5.1.2.1 through 5.5.1.2.3.

5.5.1.2.1 Short term (internal surfaces). For petroleum or PAO oil system components, short-term preservation of internal surfaces shall be accomplished with fluids in accordance with MIL-PRF-6083. For phosphate ester system components, internal surfaces shall be preserved with fluids in accordance with MIL-H-19457, with addition of a VCI in accordance with A-A-59441 per the VCI manufacturer's instructions. Water-based fluid system components in long-term storage shall require annual treatments with the system fluid, as specified in MIL-H-22072. Preservation with coatings in accordance with MIL-PRF-6083 or MIL-H-19457 fluids is satisfactory for 1 year, after which recoating is required. Since these fluids, in small quantities, are compatible with the system operating fluids, draining of residual fluid prior to installation will provide effective removal of the preservatives. See 5.9 for disposal instructions for these fluids and see 5.6.1 for definition of acceptable liquid contamination. Complete removal and degreasing is required when brazing or welding is necessary to reassemble the system. A tag with the following information shall be attached to the assembly: "The interior of this item has been (filled, coated) with corrosion preventive fluid to (specification) on (date). Coatings require replacement 1 year from date. Prior to assembly, complete removal and degreasing is required if brazing or welding is necessary to assemble components."

5.5.1.2.2 Long term (internal surfaces). For petroleum or PAO oil system components, which are expected to be in storage for considerably longer than 1 year and are easily disassembled and cleaned, the internal surfaces shall be treated with a preservative in accordance with MIL-PRF-16173, grade 2. A tag with the following information shall be attached to the assembly: "The interior of this item is coated with corrosion preventive compound, grade 2 of MIL-PRF-16173. This compound shall be removed just prior to installation." Phosphate ester system components in long-term storage will require annual treatments with fluid in accordance with MIL-H-19457, with addition of a VCI in accordance with A-A-59441, per the VCI manufacturer's instructions. The assembly shall be tagged as specified in 5.5.1.2.1.

5.5.1.2.3 External surfaces. For external surfaces of petroleum or PAO oil and phosphate ester fluid system components requiring preservation, coatings in accordance with MIL-PRF-16173, grade 1 or 4 shall be applied where the preservative will not have to be removed for system operation, or where preservative removal by scraping or solvent action would not damage the part or equipment. Where removal of the preservative will be required (such as when the preservative would otherwise be in contact with the system fluid) a coating in accordance with MIL-PRF-16173, grade 1 or 4 shall be applied. A tag with the following information shall be attached to the assembly: "The exterior of this item has been coated with corrosion preventive compound to (specification), (coating type or grade). Removal of this compound (is, is not) necessary prior to installation."

5.5.1.3 Preservative removal. The coatings in accordance with MIL-PRF-16173, grade 1 or 4 and fluid films in accordance with MIL-PRF-6083 may be removed with dry cleaning solvent in accordance with MIL-PRF-680, type II. Surfaces cleaned with solvent will be left unprotected and shall be immediately wetted with the system fluid unless welding or brazing will be required to reassemble the system. For preserved petroleum or PAO fluid system components which will not require degreasing, the system oil may be used to remove the preservative. When solvent in accordance with MIL-PRF-680 is used to remove protective coatings, complete removal of the solvent is required to avoid subsequent contamination of the system operating fluid. See 5.9 for disposal instructions for these fluids.

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5.6 System flushing. Installed piping shall be flushed before being put into operation. It is important for designers of hydraulic systems to plan for system flushing in the design phase. Dead ends without circulation shall be avoided. If there is a risk of particulate contamination moving from the dead end to the rest of the system, then the dead end shall be capable of being outflushed or a flush path loop using temporary jumpers shall be established. Flushing fluid shall be the system operating fluid or a fluid compatible with system fluid as specified by the Technical Authority directly or through applicable Navy documents. The primary difference between flushing oils and the system fluid is viscosity; lower viscosity oils (such as MIL-PRF-83282 and MIL-PRF-17672, symbol 2075-T-H) require much lower flow rates to maintain the Reynolds number specified in 5.6.1 while having much lower pressure drops than the higher viscosity oils (such as 2135-T-H in accordance with MIL-PRF-17672, 2190-S in accordance with MIL-DTL-32353, and 2190 TEP in accordance with MIL-PRF-17331). This means that by using low viscosity oils as the flushing fluid, longer pipe lengths for any given pipe size can be flushed with a given flushing plant or the same length of pipe can be flushed at higher flow rates than with high viscosity oils. Flushing shall be performed on the entire system but may be accomplished piecemeal on those circuits which can be independently cleaned and isolated. Complex systems may be divided into parts in order to assure the required flow rate in every part of the system during flushing. Hydraulic pumps, motors, complex valves, and devices that restrict flow or could be damaged by contaminants dislodged during flushing shall be removed from the circuit and temporary pipe, flushing blocks, tube, or hose substituted in their place. See figure 1 for an example of a typical flushing loop on a complex system, such as that on a submarine. System flushing shall be in accordance with the following:

a. A temporary filter, with a beta ratio of 75 for particles greater than 0.3 mils (7 microns), or better, shall be installed in the circuit being flushed and clean filter elements installed as necessary to keep the pressure drop across the filter within specified limits. Care shall be taken to select a filter with the optimum filtration ratio, so as to achieve the desired cleanliness level within an acceptable period of time. Filter bypass valves shall not be used. The flushing fluid shall be filtered through a filter with a beta ratio of 75 for particles greater than 0.3 mils (7 microns), or better, as the system is filled.

b. Temporary jumpers shall consist of hoses and hardware necessary to mate with the circuit being flushed and shall be rated to the operating pressure of the piping system. The hardware used to mate with the circuit being flushed typically includes hose fittings, isolation valves, pressure gages, and temperature gages.

c. Flushing shall be performed in the same direction as the normal fluid flow during system operation. When the normal system flow may be in either direction, such as in control port actuator lines, flushing these lines in both directions shall be required only when specified by the acquiring activity. Circuits having a rise in elevation in the direction of flow shall require reversing the direction of the flush to ensure that the system is clean in both directions only when specified by the acquiring activity. When flushing is done in both directions, flushing in the direction of normal flow shall be done last.

d. Large components that are assembled or disassembled in place to support new construction or overhaul of hydraulic systems can be cleaned in place. Applicable equipment technical manuals, drawings, and component maintenance standards, in addition to best shop practice and judgment, shall be used in determining the extent of assembly or disassembly of these large components in obtaining a visually clean condition of the component's internal parts as an alternate to flushing the component.

e. Components such as reservoirs, accumulators, and cylinders shall be flush rinsed with a high-velocity jet stream or wiped clean with a lint-free cloth as conditions permit to attain a visually clean condition. Pistons shall be removed from cylinders and accumulators for shop cleaning when practical. When wiping is necessary, lint-free toweling shall be used. Components cleaned in the shop prior to installation and properly preserved, sealed, and stored may be connected to the piping after the piping is flushed, without further cleaning (see 5.5.1.2).

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f. Unless otherwise specified (see 6.2), flushing shall continue until the fluid indicates that the class 9 cleanliness requirement of SAE AS4059 (or ISO 4406) has been achieved to suit the turbulent flow requirements of 5.6.1 and the minimum flushing time specified in 5.6.1.1. Prior to termination of flushing the system piping, the water content of the flushing fluid shall be determined when the flushing fluid is not water based. Unless otherwise specified (see 6.2), the water content shall not exceed 0.05 percent by volume for any single sample. Unless otherwise specified (see 6.2), the average of the samples taken prior to terminating the flush shall not exceed 0.03 percent by volume for any system or circuit thereof. After flushing and reassembly, the system shall be filled with fluid which meets the applicable system fluid specification, satisfies the required system fluid cleanliness requirements, and has been final filtered through a filter with a beta ratio of 75 for particles greater than 0.3 mils (7 microns), or better, during the system fill process. MIL-HDBK-407 is recommended for additional background information when planning the flushing of a system.

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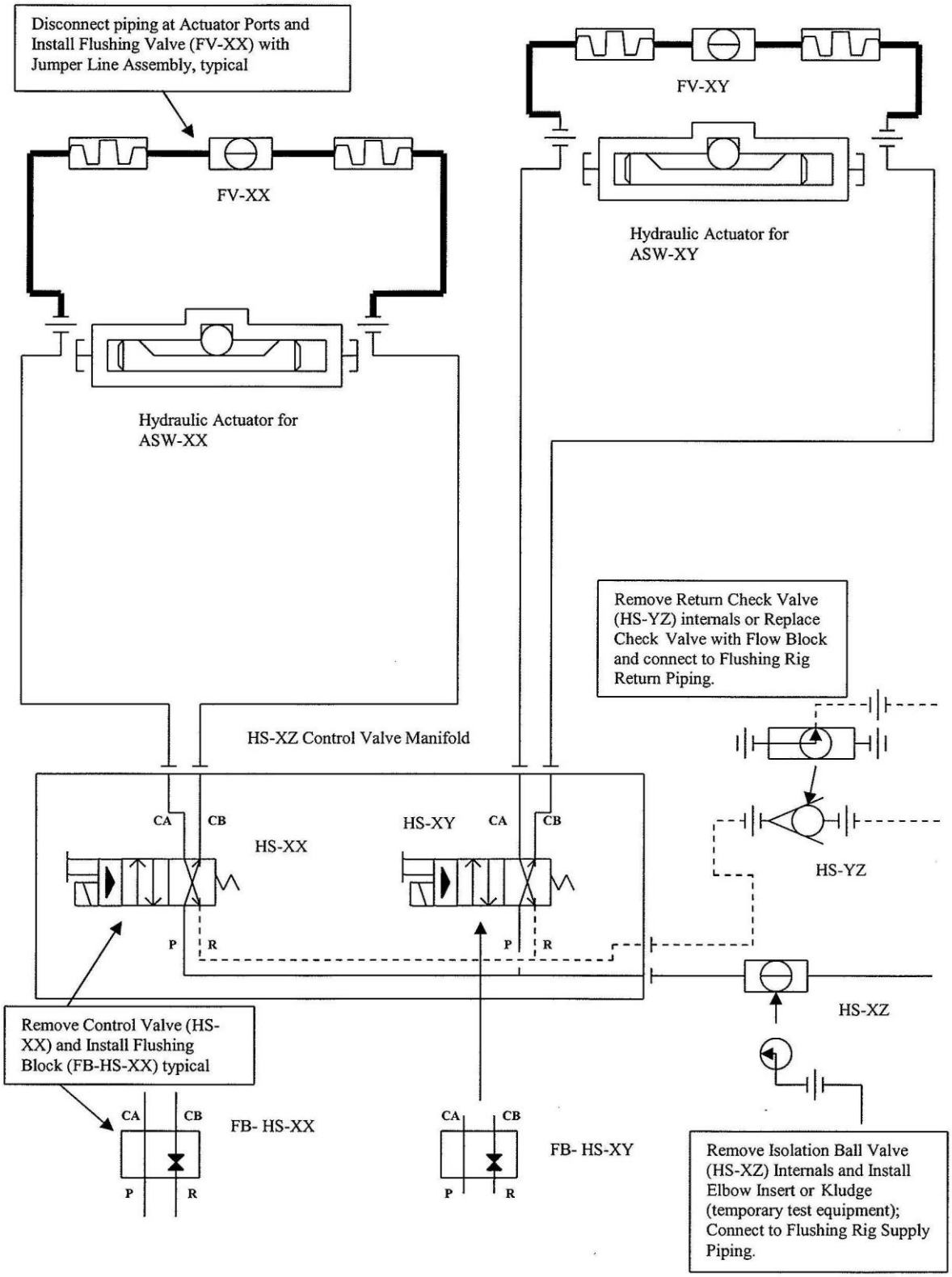


FIGURE 1. Typical flushing loop.

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5.6.1 Procedure for system operating fluid flush. Flushing with system operating fluid or a fluid compatible with system fluid as specified by the Technical Authority directly or through applicable Navy documents shall be performed. For systems utilizing MIL-PRF-17331, MIL-PRF-17672, or PAO-based fluids such as MIL-PRF-83282 or MIL-DTL-32353, flushing may be conducted with lower viscosity MIL-PRF-17672 or MIL-PRF-6083 fluids. In order to effectively remove particulate contamination from hydraulic piping, the flushing fluid should achieve turbulent flow. The Reynolds number (Re) shall be calculated using the following equation:

Re = 3,160*Q/(d*v)

Where:

Q = flow rate (gpm)

d = waterway cross-section, pipe internal diameter (inches)

v = fluid viscosity (centistokes [cSt])

Note: The value 3,160 is a constant corresponding to units of gpm, inches, and cSt.

When using metric measurements, the Re shall be calculated using the following equation:

Re = 2,120*Q/(d*v)

Where:

Q = flow rate (L/min)

d = waterway cross-section, pipe internal diameter (centimeters)

v = fluid viscosity (centipoise [cP])

Note: The value 2,120 is an approximate constant corresponding to L/min, centimeters, and centipoise.

For pipe sizes greater than ½ inch (1.3 cm), the fluid, flow rate, and fluid temperature shall be defined to achieve a Reynolds number of at least 4,000. For ½-inch (1.3 cm) pipe, if a Reynolds number of 4,000 cannot be achieved, then the pipe shall be agitated, the original established flushing time doubled, and a minimum Reynolds number of 2,000 achieved. For pipe sizes less than ½ inch (1.3 cm), when piping cannot be reasonably flushed in short enough sections to stay within the pressure limits of the piping (and system), the Reynolds number shall not drop below 2,000. Upon completion of flushing with fluids other than the system operating fluid, the system shall be completely drained and blown free of flushing fluid, and the system refilled with operating fluid. Depending upon the complexity of the system and the amount of flushing oil left behind, a second or third drain-and-fill sequence may be required to provide minimal acceptable liquid (not solid) contamination.

5.6.1.1 Minimum flushing time. Flushing at turbulent flow shall continue until the volume of piping being cleaned, or the individual flushing loop as shown on figure 1, has been displaced a minimum of 20 times, as calculated in accordance with ISO 23309 below, or 20 minutes, whichever is greater.

T = 20*V/Q

Where:

T = minutes

V = volume of piping being cleaned (gal [or liters])

Q = flow rate through the loop (gpm [or L/min])

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5.6.2 Flushing fluid samples. Unless otherwise specified (see 6.2), samples shall be taken from the bottom side of the pipe within 5 feet (1.5 m) of the inlet of the flushing filter during full flow. The size of each sample drawn shall not exceed 0.3 gal (1 liter) and shall include the fluid and contaminants trapped by the valve and adapter during the preceding 5-minute flushing period. The valve shall be opened to draw an amount greater than the volume of branch pipe and valve; the valve shall then be closed, and the sample shall be drawn after 5 minutes. The entrance to the sampling port shall be flush with the inner circumference of the pipe, and the valve connector (adapter) shall not extend into the pipe beyond the inner surface of the pipe. For taking samples from the system during flushing, diaphragm (packless) valves are recommended. Where the hydraulic system is equipped with sampling valves, samples may be taken from those valves in lieu of the location specified above. The use of in-line fluid cleanliness monitoring, such as automatic particle counters, may be used to indicate the real time cleanliness level of the flushing fluid. When using in-line or continuous fluid cleanliness monitoring equipment, a periodic verification of equipment performance by an independent fluid sample analysis shall be performed. If this analysis indicates unsatisfactory equipment performance, then the flushing activity shall take the following corrective actions:

a. Restore equipment performance.

b. Provide assurance of system cleanliness, to include repeat flushing of any suspect flow loops.

5.6.2.1 Sample analyses. The test method for determination of particulate contamination shall be in accordance with SAE AS598 or an automatic particle counter calibrated in accordance with ISO 11171. Directions in the manual for the respective instrument shall be followed. Water content shall be determined in accordance with ASTM D6304, or with commercially available automatic water-analyzing equipment that shall accurately determine water content down to 0.01 percent.

5.7 Capping and sealing open ends. Open ends shall be capped or sealed to preserve cleanliness of assemblies for stowage, transit, or installation operations. Caps shall have the same grade of cleanliness as the pipe or assembly to be sealed. The use of electrical plastic tape or other compatible tape is allowed to hold caps in place. However, the use of any tape as a cap is prohibited. Caps shall be made in accordance with NAS 847.

5.7.1 Petroleum or PAO-based hydraulic fluid caps. In systems using petroleum or PAO-based hydraulic fluid, plastic or rubber material may be used for caps. Nitrile rubber is recommended for caps. Butyl or ethylene propylene rubber, and any other rubber incompatible with petroleum or PAO-based fluids, shall not be used. The pipe end shall be completely covered with sheet plastic or rubber, and secured to the pipe outside diameter with electrical plastic tape or some other compatible tape. The plastic or rubber materials may be used for threaded and unthreaded closures.

5.7.2 Phosphate ester-based hydraulic fluid caps. In systems using phosphate ester-type fire-resistant hydraulic fluid, butyl, or other compatible rubber and compatible plastics may be used for caps. The pipe end shall be completely covered with compatible sheet plastic or butyl rubber and secured to the pipe outside diameter with a compatible tape.

5.8 Ferrous systems (except corrosion-resistant steel). When specified (see 6.2), ferrous piping, tubing, and fittings shall be phosphate coated, prior to installation, in accordance with the specified type and class of MIL-DTL-16232.

5.9 Disposal of waste materials used in cleaning. The following materials and solutions containing these materials shall not be discharged overboard. They shall be retained in containers and held for shore disposal:

a. Sodium chromate

b. Sodium dichromate

c. Lubricating oil

d. Cleaning oil

e. Corrosion preventive, solvent cutback, cold application

f. Dry cleaning solvent

g. Hydraulic fluids

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Other materials shall be disposed of in accordance with local, state, and Federal laws or retained in containers for shore disposal.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard covers basic guidance and methods for cleaning, descaling, passivating, protecting, and preserving nonferrous and ferrous metal and alloyed pipe, tubing, and fittings prior to installation. It also covers the performance of hydrostatic tests and flushing of installed systems that are used in hydraulic power transmission applications. This standard ensures appropriate and effective procedures are used to support sensitive military piping system applications, while continuing to utilize industry best practices.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. The required system cleanliness level (based on the cleanliness methods in accordance with SAE AS4059 or ISO 4406) and the minimum flushing time, as necessary (see 5.6.f).

c. The maximum water content by volume for any single sample, as necessary (see 5.6.f).

d. The maximum average water content by volume for all samples, as necessary (see 5.6.f).

e. Flushing fluid sample locations, as necessary (see 5.6.2).

f. Ferrous piping, tubing, and fittings to be phosphate coated, as necessary (see 5.8).

6.3 Subject term (key word) listing.

Hydrostatic test

Long term (internal surfaces)

Short term (internal surfaces)

Sodium dichromate

System flushing

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 4730-2021-040)

Review activities:
Army – MI
Navy – MC, MS, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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